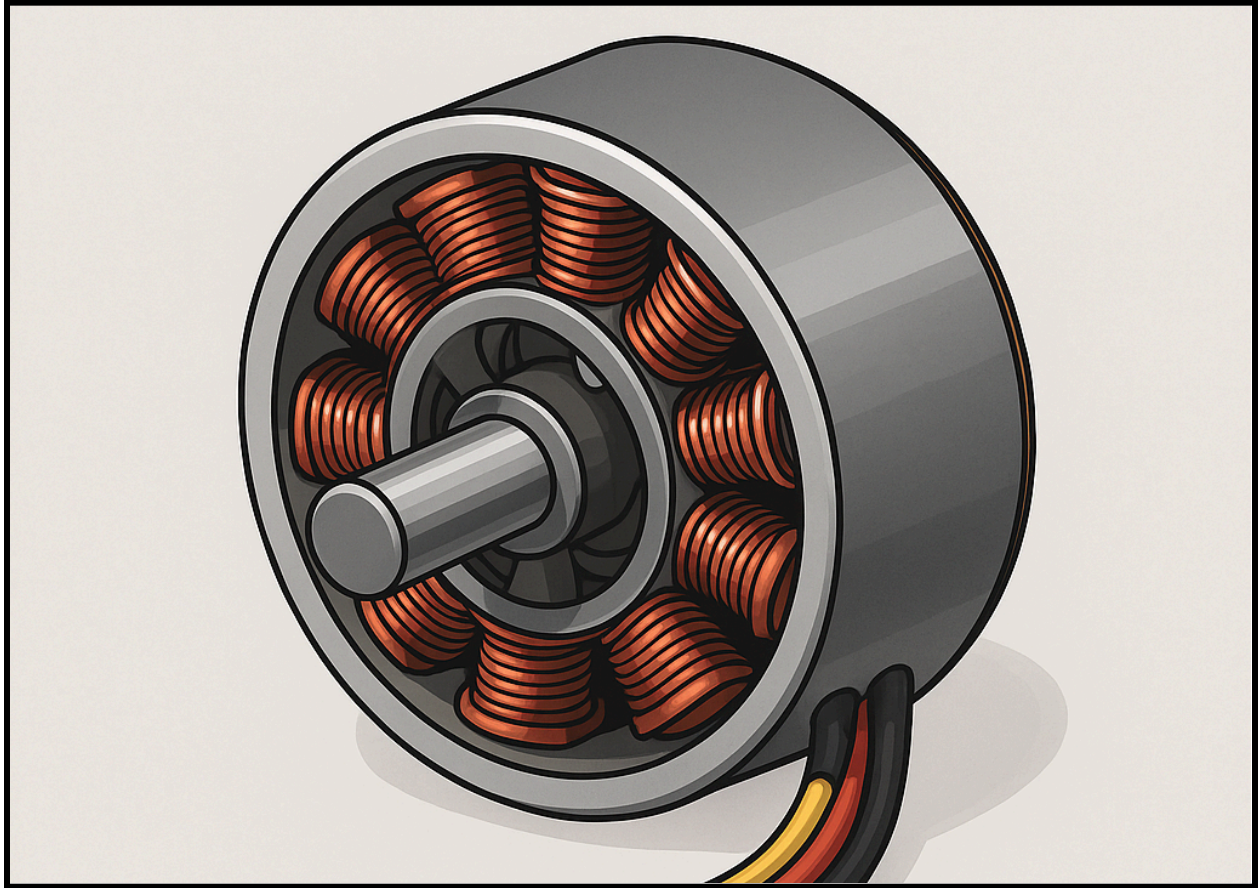


Brushless Motors

MCTR 422 Final Project Report



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INTRODUCTION

Brushless DC motors (BLDC motors) are essential components in modern electromechanical systems due to their high efficiency, reliability, and low maintenance requirements. They are widely used in applications such as drones, electric vehicles, robotics, and industrial automation. Unlike brushed motors, BLDC motors use electronic commutation via an external controller (ESC – Electronic Speed Controller), which allows for precise control of speed and torque without physical brushes or a mechanical commutator.

This project explores how brushless motors work, how control of the motors works and application of BLDC motors in a drone controlled by a microcontroller. The goal is to demonstrate a practical implementation of brushless motor control and calibration, highlighting both the electrical interfacing and the software logic required to drive and modulate these motors.

This report documents the theoretical background, materials used, build process, experimental results, and practical challenges encountered while implementing BLDC motor control in a low-budget, educational context.

Theory and Background

Brushless DC (BLDC) motors are a class of synchronous motors powered by direct current through an electronic speed controller (ESC) rather than mechanical brushes. Unlike brushed motors, which rely on physical commutation via carbon brushes and a commutator ring, BLDC motors achieve commutation electronically by switching current in the stator windings based on the rotor's position.

At the core of a BLDC motor are three main components: the stator, rotor, and sensor system. The stator consists of multiple windings (typically arranged in three phases A, B, and C), while the rotor contains permanent magnets arranged with alternating north and south poles. As the ESC sends timed pulses of current to the stator windings, a rotating magnetic field is generated. This field interacts with the rotor magnets, causing the rotor to spin and produce mechanical motion.

One of the major advantages of BLDC motors is their efficiency and reliability. Because there is no brush friction, they produce less heat, experience lower wear, and require minimal maintenance. Additionally, BLDC motors are capable of delivering high torque and fast response times, which makes them ideal for applications like drones, where precise speed control and weight-to-performance ratio are crucial.

The motor's rotational speed is determined by the frequency of the commutation pulses provided by the ESC. These pulses are usually controlled via Pulse Width Modulation (PWM) signals generated by a microcontroller (Arduino). The ESC interprets the width of each pulse and adjusts power delivery accordingly.

BLDC motors may be sensed or sensorless. Sensorless motors rely on the back electromotive force (back-EMF) generated in the windings to determine rotor position, while sensed motors use physical sensors (Hall effect sensors). The motors used in this project are typical drone-grade sensorless outrunners, driven by ESCs that interpret PWM commands from an Arduino board.

Understanding this control method is essential for any application involving dynamic speed control, especially in embedded systems or robotics. This project focused on building and demonstrating that control pathway from microcontroller signal to motor response.

Materials and Methods

This project aimed to understand, control, and demonstrate the operation of brushless DC (BLDC) motors using a practical, low-cost Arduino-based setup. The primary components used in the experiment are detailed below, followed by a description of the control method and wiring setup.

3.1 Materials

- Brushless DC Motor (DJI 2212 920KV and C1606)

A common outrunner motors often used in drones. Rated for 3S–4S LiPo batteries

and often paired with 30A ESCs.

- Electronic Speed Controller (30A ESC and aero 4-1esc)

A three-phase ESC's designed to drive brushless motors using PWM signals. It converts a standard RC signal into high-frequency switching to power each motor coil.

- Arduino Uno and Nano

Used to generate PWM control signals for the ESC and to read analog voltage from a potentiometer to vary throttle.

- Potentiometer (10 k Ω used in final test)

Acts as a throttle input, allowing real-time analog control of the motor's speed via a mapped voltage-to-PWM conversion.

- LiPo Battery (Ovonic 3S 11.1V 1400mAh 50C)

Supplies the necessary high current required by the motor through the ESC.

- Breadboard and Jumper Wires

Used to prototype the control circuit, connect the potentiometer, and supply regulated power between the ESC and Arduino.

3.2 Circuit Configuration

- The ESC's three motor leads were connected to the three wires of the DJI brushless motor.
- The ESC's servo signal plug was connected as follows:
 - White wire to Arduino digital pin 9

- Red wire to Arduino 5V
- Black wire to Arduino GND
- The potentiometer was connected with:
 - One side pin to Arduino 5V
 - The opposite side to Arduino GND
 - The middle (wiper) to analog pin A0

3.3 Code and PWM Control Logic

The program below manually sets the throttle high during setup and then drops it low to trigger calibration.

```
sketch_aug2a.ino
1  #include <Servo.h>
2
3  Servo esc;
4  int throttle = 0;
5
6  void setup() {
7      Serial.begin(9600);
8      esc.attach(9, 1000, 2000);
9      Serial.println("ESC Controller Ready. Enter 0-180 for throttle:");
10 }
11
12 void loop() {
13     if (Serial.available()) {
14         int val = Serial.parseInt();
15         if (val >= 0 && val <= 180) {
16             throttle = val;
17             esc.write(throttle);
18             Serial.print("Throttle set to: ");
19             Serial.println(throttle);
20         }
21     }
22 }
23
```

Output Serial Monitor X

Then an additional program was written in using the Servo library to generate PWM signals. The code reads the analog input from the potentiometer and maps it to a PWM output between 0 and 180 degrees, which corresponds to the signal range expected by the ESC (typically 1000–2000 μ s). A simplified version of the code is shown below:

```
sketch_aug2a.ino
1  #include <Servo.h>
2
3  Servo ESC;
4  int Speed;
5
6  void setup() {
7      ESC.attach(9, 1000, 2000);
8  }
9
10 void loop() {
11     Speed = analogRead(A0);
12     Speed = map(Speed, 0, 1023, 0, 180);
13     ESC.write(Speed);
14 }
15
```

3.4 Troubleshooting and Motor Response

Initial issues included incorrect throttle mapping, poor ESC calibration, and a potentiometer that was too low in resistance. Once corrected with a higher-resistance potentiometer and stable wiring, the system responded reliably. The motor would spin when the potentiometer reached a threshold (around 160 on the PWM scale), with speed varying according to potentiometer input. The Arduino was observed to remain powered from the ESC's 5V rail when USB was unplugged.

Results

The project successfully demonstrated basic operation and control of brushless DC (BLDC) motors using an Arduino Uno and Electronic Speed Controllers (ESCs). After initial difficulties with ESC calibration, signal timing, and power delivery, the motors were ultimately able to respond to throttle commands sent via PWM signals generated by the Arduino.

During testing, the ESC emitted the expected initialization beeps. After calibration routines were applied, a dedicated calibration control code was uploaded to the Arduino. Once the proper procedure of bringing the throttle (via potentiometer) to maximum and then down to minimum was followed, the motors began responding as expected. Early test results showed short bursts of motor activity, which confirmed that PWM control was functioning. Once calibration was completed correctly, consistent motor spin was achieved in response to throttle adjustments, validating both the code and electrical setup.

The ESCs were calibrated manually by applying maximum throttle at power-up and reducing to minimum after the initial beep sequence. This process established the throttle range, and further tuning allowed for real-time motor speed adjustments via analog input.

Ultimately, the project proved that an Arduino-based setup can effectively interface with brushless motors using ESCs, providing the foundational functionality required for flight control systems. The results support a working understanding of how brushless motors are powered, how ESCs translate PWM signals into phase-controlled commutation, and how manual calibration procedures affect motor response.

Overview

This project provided valuable insight into the core principles behind brushless DC motor operation, ESC control, and Arduino-based signal generation. The experience of integrating and calibrating brushless motors manually highlighted the complexity behind even basic electromechanical control.

One major challenge encountered was ESC calibration. Brushless ESCs are designed to receive specific PWM ranges during startup and require tightly timed throttle signals to properly enter calibration mode. Since most ESCs expect signals from commercial flight

controllers, reproducing these signals manually via Arduino required careful attention to signal timing and power sequencing. Even slight delays or out-of-order steps in the calibration routine often led to repeated error beeps or no motor response.

Another complication was the lack of real-time feedback or closed-loop control. Without feedback from sensors like gyroscopes or encoders, the system relied entirely on open-loop PWM commands. This made precise speed control difficult and limited the system's ability to stabilize or sustain motion under varying loads or conditions.

Additionally, working within a limited budget forced creative problem-solving. I had to purchase two sets of motors and eventually a 4-1 esc controller to develop a working prototype that could demonstrate the core principles of ESC control. The fact that the system was able to deliver a motor response and react to throttle input confirmed that a basic understanding of motor control, timing, and signal generation had been achieved.

Overall, this hands-on experience solidified key engineering skills, such as reading datasheets, troubleshooting inconsistent hardware behavior, and understanding the interaction between software-generated signals and hardware-level electrical components. These are directly applicable to both robotics and embedded systems applications, which are relevant to many career paths in mechatronics and electrical engineering.

Conclusion

The project revealed just how precise and sensitive brushless motor systems are to control signals, and underscored the importance of timing, hardware compatibility, and signal fidelity. Despite setbacks, including signal mismatches, calibration difficulties, and component limitations, a working demonstration was ultimately achieved through persistence and iterative testing.

Most importantly, the project showed a practical understanding of how brushless motors function and how they are driven in real-world applications. These motors are the backbone of modern drones and many robotic systems, and being able to manually control one from scratch provided hands-on insight into a critical technology.

During testing and prototyping to showcase the applications of the brushless motors(in this case a drone), I found greater success using the C1606 motors in combination with a

4-in-1 ESC module. This setup proved more stable and manageable than my earlier configuration, which used DJI 2212 920KV high performance motors paired with individual ESCs. The initial setup presented a number of compatibility and calibration challenges that ultimately hindered consistent motor response. In contrast, the 4-in-1 ESC offered simplified wiring, better integration, and more reliable signal control, resulting in a cleaner build and more responsive performance overall.

Because there is a direct correlation between my time in engineering school and how poor I am, as part of the planning for this project, I selected a drone frame and components that could be repurposed or expanded upon for future engineering applications. In particular, I opted for a modular frame, a 4-in-1 ESC along with individual esc's, and two sets of brushless motors to provide flexibility. While this project focused on demonstrating a practical understanding of brushless DC motor operation and control, the drone served as a widely recognized application platform for brushless motors. Looking forward, I hope to extend this work into a more advanced and unconventional design; potentially exploring the feasibility of a lighter-than-air drone platform as a senior design or personal capstone project. This approach will hopefully manage cost and build time, and ideally positions me to continue learning and innovating beyond this initial prototype.

This project forms the basis for future work in embedded control, autonomous systems, and robotics. The skills gained, signal generation, motor control, troubleshooting embedded systems, will be directly applicable in future coursework and professional engineering projects. While budget and time constraints created challenges the foundational understanding achieved here is a meaningful step toward more advanced designs and career-level engineering proficiency.

References

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2. Mohan, Ned. Electric Machines and Drives: A First Course, Wiley, 2012.
3. Arduino Documentation — www.arduino.cc

4. Robotic Controls, ESC Calibration Tutorials.
5. Datasheets and specifications for DJI 2212 920KV and C1606b brushless motors and associated ESCs.
6. MPU-6050 Gyro/Accelerometer Module Documentation.
7. Various community forums, including Arduino and drone DIY forums, for troubleshooting calibration, ESC timing, and PWM control.